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Inventor(s)

Daffer; Steven J.

ENVIRONMENTALLY CONTROLLED MULTI-MODALITY CHAMBER, COMMUNICATION SYSTEMS AND METHODS OF USE

Abstract

A method of controlling an environmentally controlled chamber includes providing a chamber with at least one infrared heater, a screen configured to stream content, the screen positioned behind an insulating barrier comprising a substantially optically transparent portion, a camera configured to record a user and speakers within the chamber. The method includes sending the user's physiological and/or psychological data and environmental conditions within the chamber to a location remote from the chamber for monitoring, and changing environmental conditions within the chamber based upon the received physiological and/or psychological data or sensed environmental conditions from the location remote from the chamber.

Inventors: Daffer; Steven J. (Edina, MN)

Applicant: Visibelle Derma Institute, Inc. (Bloomington, MN)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] The present application is based on and claims the benefit of U.S. provisional patent application Ser. No. 63/556,946, filed Feb. 23, 2024, the content of which is hereby incorporated by reference in its entirety.

BACKGROUND

[0002] The present disclosure relates to a communication and control system for use with an environmentally adjustable chamber such that the environment within the chamber can be modified for one or more users within the chamber. More particularly, the communication and control system for use with at least one environmentally adjustable chamber allows an environment to be adjusted in real time while the communication system allows an observer at a location remote from the chamber to communicate with the one or more persons conducting an activity within the chamber in real time.

[0003] Many people want to exercise, train, do physical therapy and/or rehabilitation activities at sub-ambient or elevated temperatures for the enhanced results relative to conducting the same activities at ambient temperatures. Exercising, training, and/or conducting physical therapy and rehabilitation at sub-ambient or elevated temperatures has many benefits. Some non-limiting benefits include increasing the rate at which calories are consumed and increasing and decreasing body temperature and stresses on the body, both of which increase a person's performance compared to conducting the same activities at ambient conditions. Additionally, exercising, training, and/or conducting physical therapy and rehabilitation at sub-ambient or elevated temperatures can increase cardiovascular health and stamina and may allow a person to achieve a greater range of motion to increase flexibility.

[0004] There is a need to customize an environment for people conducting activities including, but not limited to, elevating temperatures above ambient temperatures and/or below ambient temperatures for one or more users. There is also a need to customize a physical activity regime for the one or more users by receiving real time feedback while performing the activity within the chamber. There is also a need for an observer to be remote from the controlled environment chamber while providing real time feedback to the user, as the observer cannot withstand prolonged times within the controlled environment chamber due to the physical demands on the body.

SUMMARY

[0005] An aspect of the present disclosure relates a method of controlling an environmentally controllable chamber. The method includes providing a chamber with at least one infrared heater, a screen configured to stream content, the screen positioned behind an insulating barrier comprising a substantially optically transparent portion, a camera configured to record a user and speakers within the chamber. The method includes sending the user's physiological and/or psychological data and environmental conditions within the chamber to a location remote from the chamber for monitoring, and changing environmental conditions within the chamber based upon the received physiological and/or psychological data or sensed environmental conditions from the location remote from the chamber.

[0006] Another aspect of the present disclosure includes a method of controlling a plurality of environmentally controlled chambers. The method includes providing a plurality of spaced apart chamber with at least one infrared heater, a screen configured to stream content, the screen

positioned behind an insulating barrier comprising a substantially optically transparent portion, a camera configured to record a user and speakers within the chamber. The method includes sending the user's physiological and/or psychological data and environmental conditions within each of the plurality of chambers to a location remote from the chamber for monitoring. The method also includes changing environmental conditions within one of the plurality of chambers based upon the received physiological and/or psychological data or sensed environmental conditions within the one chamber of the plurality of chambers based upon signals sent from the location remote from the one chamber of the plurality of chambers.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. 1 is a cutaway perspective view of a chamber with a portion of a control system.

[0008] FIG. 1A is an enlarged view of a recess configured to retain a hand held portable electronic device.

[0009] FIG. 2 is a cutaway perspective view of the chamber with another portion of the control system.

[0010] FIG. 3 is a cutaway perspective view of the chamber with yet another portion of the control system.

[0011] FIG. 4 is a schematic view of a chamber with controllable environmental modalities.

[0012] FIG. 5 is a schematic view of an observer's equipment.

[0013] FIG. 6 is an exemplary graphic user interface for the observer's use.

[0014] FIG. 7 is a schematic view of a control system for monitoring and controlling chambers from a common facility.

[0015] FIG. 8 is a schematic view of a control system with a plurality of observers located in a separate facility from the chambers.

[0016] FIG. 9 is a schematic view of a control system for residential use.

[0017] FIG. 10 is a schematic view of a control system with an observer located in a facility remote from users in a plurality of facilities.

DEFINITIONS

[0018] An “user” as used herein is a person that is using a chamber with a customizable environment. Non-limiting activities that the user can be doing include actively exercising for cardio vascular condition, strength training, speed and agility training or flexibility training. The user can also be passively using the chamber by sitting, standing, laying prone and/or for relaxation or meditation purposes. The user can also be using the chamber for physical rehabilitation purposes.

[0019] An “observer” as used herein is a person that is located in a position that is spaced from the chamber where the observer interacts with one or more users in one or more chambers. The observer can be a physical trainer who is working with one or more users to maximize physical benefits from exercising in a chamber, a physical therapist or other medical professional or a person that monitors the user while the user is in the chamber. The observer can monitor one or more users, the one or more user's physiological data, and/or environmental conditions within the one or more chambers. Based upon the received inputs regarding the conditions in the one or more chambers and the one or more user's physiological data and/or psychological data, the observer either adjust environmental conditions in the one or more chambers based upon the physiological data of the one or more users and/or instruct modification of the one or more user's activities. The observer can also communicate with the one or more users in the one or more chambers utilizing one-way or two-way communication.

[0020] A “control system” as used herein includes a communication system and a system for

controlling and adjusting one or more environmental conditions in the one or more chambers. The control system is configured to control the environmental conditions in each of the one or more chambers while allowing for communication between an observer and a user independent from the other one or more chambers.

[0021] A “one-way” communication system within the control system as used herein is a system that allows one or more observers to view a video stream of one or more users in one or more environmentally controllable chambers where the one or more observers can select to audibly communicate with one or more selected users in one or more environmentally controllable chambers independent from the other chambers.

[0022] A “two-way” communication system within the control system as used herein is a system that allows one or more observers and the one or more users to send and receive video streams of each other and where the observer can select to audibly communicate with one or selected users in one or more chambers. An alternative two-way communication allows the user and the observer to communicate in real time while the observer receives a video stream of the user. Another two-way communication systems allows video streams of the user and observer to be viewed by the other person and allows the user and observer to communicate in real time.

[0023] A “wearable device” as used herein is a device that is worn by a user that is configured to monitor or sense physiological data of the user and to transmit the monitored or sensed physiological data to the control system.

[0024] “Physiological and/or psychological data” as used herein includes, but not limited to, internal body temperature, pulse rate, heart rate, blood pressure, blood oxygen content, eye movement, surface skin temperature, range of motion, strength output, speed of movement, speed of reaction, repetition of movement, maximum rate of oxygen consumption that the body is able to use (VO.sub.2 max) and distance moved during exercise.

[0025] “Environmental conditions” as used herein includes, but is not limited to, temperature, humidity, salt and mineral content, visible light wavelength, ultraviolet wavelength, infrared wavelength, biomagnetism, pulsed electromagnetism, air speed and volumetric flow rate, oxygen content, sound, ultrasound waves, radiofrequency waves, visual content and/or holistically enhanced air and/or infused aroma in the chamber.

DETAILED DESCRIPTION

[0026] The present disclosure relates to at least one chamber that is configured to have a customizable environment, including an elevated or sub-ambient temperature, and a one-way or two-way communication system within a control system associated with the at least one chamber. The control system allows one or more observers to view one or more users in one or more chambers and provide at least audible feedback in real time to the user and/or adjust environmental conditions within the one or more chambers from a location remote from the one or more chambers. The combination of the one or more chambers and the control system allows the environmental conditions within the one or more chamber to be monitored and adjusted, while allowing the one or more users to receive real time feedback or instruction from the one or more observers that is located remote from the one or more chamber. Additionally, the one or more observers can visually monitor the one or more users and/or the one or more user's physiological data and/or psychological data such that the results of activity in the one or more environmentally controlled chambers can be more effective.

[0027] The present disclosure also provides components of the communication system within the control system within each of the one or more chambers that allows the one or more users to conduct the selected activity, whether active or passive, while being viewed by one or more observers in real time, where the one or more observers can provide feedback in real time to the user from a location remote from the chamber. The communications system within the control system includes at least a camera that is configured to transmit a video stream of the user within the chamber to a screen that is viewed by the observer at a location remote from the

chamber in real time. The communications system within the control system also includes at least a speaker in the chamber that is configured to provide audible feedback spoken into a microphone by the observer at the location remote from the chamber and transmitted through the communication system within the control system to the user within the chamber in real time. The configuration of the camera and speaker in the chamber and the screen and microphone remote from the chamber is a one-way communication from the observer to the user, as the observer can provide feedback to the user while the user cannot communicate with the observer or trainer.

[0028] In another embodiment, the chamber includes the camera and the speaker, as well as a microphone and an optional screen. The observer also has a speaker and a camera along with the microphone and the screen. With each party having access to a camera, a speaker, a microphone and a screen, the user and the observer can see and converse with each other in real time, making the chamber and/or the exercise experience more interactive. The configuration of the control system where each party has a screen, a microphone, a video camera and a speaker is a two-way communication between the observer or trainer and the user. Another two-way communication system is a configuration where the observer and user can communicate in real time while the observer receives a video stream of the user.

[0029] In some embodiments, the screen, microphone, video camera and speaker are individual components. In other embodiments components can be combined into one or more pieces of equipment. In some embodiments, the equipment is contained within a smart television, having a camera and microphone along with the screen and speakers.

[0030] In some embodiments, the observer is located within the same space at a location of a chamber where one-way or two-way communication in the control system is utilized between the user in the chamber and the observer. In another embodiment, the observer is located within the same space at a location of a plurality of chambers wherein the control system allows the observer to communicate with one or more users in each of the plurality of chambers in real time with either the one-way communication or the two-way communication. In some embodiments, the observer can choose between the one-way and the two-way communication system for each of the chambers.

[0031] In other embodiments, the observer is located in a facility that is remote from one or more the chambers in another facility. The control system between the two facilities allows the observer to communicate with the user using a one-way communication or two-way communication between the observer and the user(s) in the one or more chambers in real time. In some embodiments, the observer can choose between the one-way and the two-way communication system.

[0032] In another embodiment, the observer is located in a facility that is remote from two or more chambers in two or more facilities, which are also remote from each other. The control system allows the observer to communicate with the two or more users in the two or more chambers located in the two or more remote facilities utilizing one-way or two-way communication with each other in real time. In some embodiments, the observer can choose between the one-way and the two-way communication system.

[0033] The control system also includes a monitoring and control capabilities where the system can monitor environmental conditions within the chamber and/or physiological data of the one or more users while the chamber is heated with infrared heaters or cooled with chilled air. The observer can utilize the collected environmental and physiological and/or psychological data to optimize the user's benefits from activities being conducted within the chamber by changing environmental conditions. By way of non-limiting example, if the user's internal body temperature is overly elevated, the observer can cause the temperature within the chamber to decrease by reducing the heat input or causing chilled air to be forced into the chamber. Alternatively, if the user is not achieving the desired effects of the activity in the chamber, the temperature can be increased by changing the heat output from the heaters, bringing additional heaters on-line or forcing hot air into

the chamber. Also, by way of non-limiting example, if the blood oxygen content of the user is too low, oxygen enriched air can be directed into the chamber to address the blood oxygen content of the user. Conversely, the oxygen concentration in the chamber can be reduced to simulate altitude for oxygen acclimation. In other embodiment, air ionizers and/or air purifiers can be manipulated by the control system. In yet other embodiments, a humidity/steam system or an aromatherapy system can be manipulated by the control system to change the environmental conditions within the chamber. In yet other embodiments, visible light, such as but not limited to, controlled LED light, and/or electromagnetic energy, such as but not limited to, pulsed electromagnetic energy, can be manipulated by the control system to change the environmental conditions within the chamber. It is within the scope of the present disclosure to adjust any of the environmental variables individually or in combination with other environmental variables based upon the environmental and/or physiological data to enhance the user's experience.

[0034] The present disclosure utilizes a chamber that has one or more infrared heaters that are used to heat the chamber to the selected elevated temperature, where the one or more infrared heaters emit photons. The one or more infrared heaters can emit far infrared (FIR) wavelengths, mid infrared (MIR) wavelengths and/or near infrared (NIR) wavelengths to provide different effects to the user. By way of example FIR wavelengths penetrate deeper into the body to increase core temperature while NIR wavelengths provide topical heat to the skin of the user. MIR wavelengths provide both some surface heating and some penetration into the body. In some embodiments, the infrared heaters only emit FIR wavelengths. The one or more infrared heaters can also be controlled with the control system.

[0035] However, most electrical equipment such as screens, speakers, cameras and/or microphones cannot withstand the elevated temperatures for extended periods of time. Elevated temperatures include temperatures above 130° F. The present disclosure is configured to allow the electronic equipment to be used for extended periods of time within chambers at elevated temperatures by preventing the equipment from being fully exposed to the elevated temperatures while allowing the user or user to utilize the electronic equipment. By way of non-limiting example, a video display screen can be positioned behind a tempered glass pane that allows the screen to be viewed while in the chamber without exposing the screen to excessive temperatures. Also, a compact electronic device, such as a cellular telephone or an electronic tablet, can be stored in a recess that is protected from the elevated temperatures by a movable cover, such as tempered glass, such that the user can access the electronic device when desired and then the device can be placed back into the recess to protect the device from the elevated temperatures.

[0036] A camera may be installed behind protective material, such as tempered glass, such that sensitive electronic equipment is not exposed to elevated temperatures. Finally, a speaker and/or a microphone can optionally be installed where sound can be transmitted around or through a protective barrier so that the user can hear instructions from the observer and the user can also communicate with the observer. However, in some embodiments, the speaker is able to withstand the elevated environmental conditions of the chamber.

[0037] In another embodiment, the electronic equipment within the environmental chamber can include a commercially available electronic device, such as a smart television that includes at least a camera, a microphone and optionally a speaker, and a screen capable of receiving a video stream and a speaker for transmitting an audio feed. In other embodiments a hand held device, such as a cell phone or a tablet, can be used to provide the camera, speaker, screen and microphone where the hand held device streams to a television located behind the tempered glass. In one embodiment, the hand held electronic device and the screen are hardwired into the control system. In other embodiments, the hand held electronic device and the screen are Wi-Fi enabled to communicate with the control system wirelessly.

[0038] Referring to FIG. 1, a chamber **10** includes a left wall **12**, a back wall **14**, a right wall **16** and a front wall **18**. The chamber **10** includes a floor **20** that attaches to the bottom edges of the walls

12-18 and a ceiling **22** that attaches to the top edges of the walls **12-18**. The chamber **10** includes a door **24** in the left wall **18** that allows for entry and egress from an interior space **26** within the chamber **10**. The chamber **10** includes at least one infrared heater **28** that is used to heat the interior space **26** to the selected elevated temperature. Typically, the at least one infrared heater **28** includes a far infrared spectrum (FIR) heater having wavelength ranges from about 5,600 nm to about 100,000 nm. Typically, the chamber **10** includes more than one IR heater **28** to aid in heating the interior space **26**. By way of non-limiting example, a chamber with multiple IR heaters is disclosed and illustrated in Daffer U.S. Printed Application Publication No. 2020/0384288A1, which is incorporated by reference herein that includes MIR and NIR heaters. The present disclosure can be utilized in the photonic chamber Daffer U.S. Printed Application Publication No. 2020/0384288A1 with the addition of the control system and its components and optionally removing the bench and to provide an even floor without a bench or heater. The interior space **26** is sized to allow any number of people to exercise, and is typically sized to allow 1 to 4 people to exercise therein.

[0039] The chamber **10** includes components of a communication system within a control system **40**. The components include a camera **50** is positioned through an aperture **52** in the left wall **12** where the camera **50** is at a height that provides a view of person while exercising. In some instances, the camera **50** is capable of tracking movement. However, the aperture **52** can be located in any of the other walls. The aperture **52** for the camera **50** can be covered with a substantially transparent, insulating material to protect the camera **50** from the elevated temperature within the chamber **10**. A typical, non-limiting, substantially transparent, insulating material includes a pane of tempered glass **51**.

[0040] The communication system within the control system **40** also includes one or more speakers **48** that are located in the back wall **14**. However, the speakers **48** can be located within any of the walls. The one or more speakers **48** are configured to receive signals either through a hard wire or wirelessly such that the audio stream from the observer can be heard. The camera **50** and the one or more speakers **48** are used for one-way communication from the observer to the user.

[0041] The communication system within the control system **40** includes a screen **42** within a recess or cut out **44** within the left wall **46** that is covered by a pane of tempered glass **47** to shield the screen **42** from the elevated temperatures within the interior space **26**. The screen **42** is configured to receive signals either through a hard wire or wirelessly from an external router such that the observer can be viewed by the user. However, the recess or cut out **44** can be located in any of the other walls.

[0042] The camera **50** can be hard wired or wirelessly connected to the communication system within the control system **40** such that a stream of the user can be viewed by a third party, such as the observer. The observer can the communicate with the user(s) via the speakers **48**. The system optionally includes a microphone **51** that is configured to send an audible stream from the user to the observer or trainer.

[0043] Referring to FIGS. **1** and **1A**, the chamber **10** can also optionally include a recess **60** sized to accept a hand held device, such as, but not limited to a cellular telephone **62**. The opening **64** to the recess is covered with a door **66** having an insulating material, such as at least partially a pane of tempered glass. A handle **68** is attached to the door **66** to move the door **66** from a first position covering the recess to a second position to provide access to the recess **60**, which also allows the user to protect the hand held communication device from excessive heat for extended periods of time when positioned in the recess **60** and the door **66** covers the recess **60**. When in the door **66** is in the second open position, the hand held device **62** can be retrieved and used for a short period of time with the chamber **10** at the elevated temperature. Once the use is completed, the hand held device **62** is placed into the recess **60** and the door **66** is placed into the first position to protect the hand held device from the elevated temperatures. The door can be positioned from the first position to the second position and back using a connecting member, such as a hinge, a slide and/or a spring biased hinge or slide. In some embodiments, the screen **42** is configured to receive and display

stream directly from the hand held device **62** such that audio and video streams received by the hand held device **62** are received and displayed or able to be heard through the screen **42**.

[0044] Referring to FIG. 2, another embodiment of the communications system within the control system within the chamber **10** is illustrated at **70**. The communication system within the control system **70** includes a recess **72** that is configured to accept a hand held electronic device **74** that includes a camera **76** configured to record the user and a speaker **78** configured to provide audible communication to the user from the observer. The device **74** can include a microphone **80** and a screen **82** where the microphone **80** allows the user to communicate with the observer and the screen **82** can be streamed to a larger screen **84** located within another recess **85**. Alternatively, the larger screen **84** can receive a video stream for the control system **70**. Both the larger screen **84** and the hand held communication device **74** are protected with an insulating and substantially optically transparent material such as, but not limited to tempered glass **86** and **88**. In some instances, the recesses include openings to cause convective cooling. In other instances, cooling fluid, such as air is forced through the recess **72** and **85** to actively cool the larger screen **84** and/or the hand held electronic device **74**.

[0045] Referring to FIG. 3, another embodiment of a communication system within a control system illustrated within the chamber is illustrated at **71**. Instead of having multiple components. The communication system with the control system **71** includes a smart television with a screen **73**, a camera **75**, speakers **77** and a microphone **79**. The smart television is located within a recess **81** that is covered by tempered glass **83** to protect the smart television from excessive heat.

[0046] Referring to FIG. 4, a schematic view of the chamber **10** with the environmental sensors within the chamber, physiological sensors and external environmental modalities used to manipulate the environment are illustrated. The chamber **10** includes the one or more IR heaters **28** and chromatic visible lights **29**, such as LED light arrays as disclosed in Daffer U.S. Printed Application Publication No. 2020/0384288A1. The chamber also includes a temperature sensor **120** to determine the temperature in the chamber, an optional an oxygen sensor **122** to determine oxygen concentration in the chamber and an optional hygrometer **124** to measure humidity in the chamber. The chamber **10** can include biomagnetic emitting materials **27**, such as but not limited to steel, cobalt and nickel, and/or electromagnetic pulse engines **25** configured to emit pulses of electromagnetic energy.

[0047] External environmental modalities include an air cooler or chiller **130**, an air heater **132**, an oxygen generator **134**, an aroma therapy nebulizer or dispenser **136** and/or a humidity or steam generator **138**. The environmental modalities also include a fresh air intake **140** and an exhaust system **142**. Internal modalities within the chamber include an air ionizer **145** and/or an air disinfectant **147**, such as a UV light source.

[0048] The user is in communication with at least one wearable device **144** that measure one or more of the physiological data. The one or more wearable devices **144** and the temperature sensor **120** and optional oxygen sensor **122** and hygrometer **124** are in communication with the control system **40**, **70** and/or **71** such that the observer can monitor the environmental conditions within the chamber **10** and the physiological data of the user.

[0049] Based upon the analyzed data from one or more of the wearable device **144**, the temperature sensor **120**, the oxygen sensor **122** and/or the hygrometer **124**, the user can manipulate of the modalities by either energizing the modality **130**, **132**, **134**, **136**, **138**, **140**, **142**, **145** and/or **147** manipulating control valve **131**, **133**, **135**, **137**, **139**, **141** and **143** for the modalities **130**, **132**, **134**, **136**, **138**, **140** and/or **142**, all respectively.

[0050] By way of example if a user's internal body temperature is below a predetermined range for optimal hypothermic conditioning, the air heater **132** may be energized and the control valve **133** may be opened using the control system **40**, **70**, **71** to force hot air into the chamber **10** to quickly raise the environmental temperature within the chamber **10**, which in turn should raise the user's

internal body temperature. Conversely, if the user's internal body temperature is too high, the observer can energize the air chiller **130** and/or open the control valve **131** using the control system **40, 70, 71** to quickly lower the environmental temperature within chamber **10**, which should lower the user's internal body temperature.

[0051] If the user's VO.sub.2 max is sensed as declining, the observer can energize the concentrated oxygen generator **134** and/or open the control valve **135** using the control system **40, 70, 71** to increase the oxygen concentration in the chamber to stabilize or increase the user's VO.sub.2 max. If the user is conditioning for exertion in a humid climate, the observer can raise the humidity level within the chamber **10**, by energizing that humidity or steam generator **138** and/or opening the control valve **138** using the control system **40, 70, 71**. When nebulizing of a topical liquid, such as a liquid vitamin and/or cosmenutraceutical and/or aroma therapy is desired in the chamber, the observer can energize a nebulizer and/or the aroma therapy dispenser **136** and/or open the control valve **137** using the control system **40, 70, 71**. When the observer desires to change out the air in the chamber **10**, the air intake **140** can be energized and/or the control valve **141** and the exhaust **142** can be energized and/or the control valve opened using the control system **40, 70, 71** to change out the air in the chamber **10**.

[0052] When the user's activity requires a screen **146**, the observer can control the screen **146** with the control system **40, 70, 71** and can provide information, such as input from a trainer when exercising, or calming visual effects, such as when meditating. Similarly, observer can control the speaker **148** with the control system **40, 70, 71** to provide information and/or calming sounds.

[0053] The observer can control the modalities by themselves or in combination with one another to adjust the environmental conditions in the chamber **10**, based upon and sensed condition, physiological data and/or time. The combinations of modalities modifications to the chamber **10** allows the user to maximize the beneficial effects of the activity in the chamber **10**.

[0054] Once, the user is finished in the chamber **10**, the UV lights **147** can be energized to emit UVC lights that is used to cleanse and sterilize the chamber air and surfaces. The UVC lights can aid in providing a more clean and sterile environments between sessions by a user or by different users using the same chamber **10**.

[0055] Referring to FIG. 5, the observer O can have additional equipment of the control system **40, 70** and **71** where the observer can have a screen **90** configured to receive the video stream from the camera **50** or **76** and a microphone **92** configured to send an audible stream to the speaker **48** or **78**. This configuration provides for one-way communication from the observer with the user.

[0056] The observer's equipment can include a camera **94** configured to stream video to the screen **42, 73** and **82/84** and a speaker **96** configured to receive an audible stream from the microphone **51, 79** or **80**. The use of the screen **90**, the microphone **92**, the camera **94** and the speaker **96** with the electrical components in the communication system **40, 70** and **71** allows for two-way communication between a person a chamber and the observer or trainer. In many instances when the observer is working with multiple chambers with multiple users, the control system **40, 70** and **71** includes a control feature **96** that allows the observer to switch between one-way and two-way communication, so that the observer can monitor all chambers with the users while directly communicating with a single chamber. Further, the observer also has a control feature **98** to only turn the audio feed from the microphone **92** to a single chamber.

[0057] Referring to FIG. 6, an exemplary graphic user interface (GUI) on the screen that is either a touch screen or controlled by a keyboard and mouse is illustrated at **100**. The GUI **100** include one or more split screen displays **102.sub.a-102n** where n is the number of chambers being monitored. Each split screen display **102.sub.a-102n** includes a display of the monitored or sensed physiological data from devices worn by the user at **104.sub.a-104n** located within each split screen **102.sub.a-102n**. Each split screen **102.sub.a-102n** includes a display of environmental conditions **106.sub.a-106n** located within each split screen **102.sub.a-102n**. Each split screen **102.sub.a-102n** includes a display **108.sub.a-108n** to activate an audio stream from the observer to the selected

chamber such that one-way or two-way audio communication can be established with one or more selected chambers. Each split screen **102.sub.a-102n** also include a display **110.sub.a-110n** to activate and deactivate a video stream of the observer to the user in a selected chamber.

[0058] In some embodiments upper and lower limits are provided to the system for the physiological or psychological data. When the monitored or sensed physiological or psychological data moves beyond the upper or lower limit, the GUI provides an indication to the observer that attention is required. In some instances, the observer will open a one-way or two-way communication to determine the cause of the indication and/or result in the observer adjusting one or more environmental conditions through the GUI **100**.

[0059] Referring to FIG. 7, a control system is illustrated at **200** that is contained within a single facility where the control system is configurable to be one way communication, two way communication or a modified two way communication where video is streamed and audio feedback is only provided to selected chambers. The control system **200** as illustrated is configured to allow an observer to be located in a location **202** remote from one or more selected chambers. In some embodiments, the location **202** is a heated chamber and in other embodiments the location **202** is in ambient conditions. As illustrated, the one or more heated chambers include six chambers **204, 206, 208, 210, 212** and **214**. Chamber **204** is designated as **n**, and all subsequent chambers **206, 208, 210, 212** and **214** are designated with the next consecutive number. It is noted that chamber **214** is designated as **n+x**, meaning the control system **200** can be utilized with any of a number of chambers and is not limited to the illustrated six chambers.

[0060] Typically, each of the chambers **202-214** are similarly equipped with a screen, camera, speakers, microphone, and a control panel and an optional a recess for storing hand-held electronic equipment. The components of the electronic equipment can be separate or can be included in one or more devices, such as a cellular telephone having all four pieces of equipment and or a television having a screen and a speaker. Having the chambers **202-214** being similarly equipped allows the chambers to be interchangeable.

[0061] The observer is in a location **202** that is remote from the chambers but in the same facility where the observer or trainer can monitor each chamber **202-214** on the screen **90**. The observer receives at least a video stream from one or more of the chambers **202-214** along with the physiological data and/or environmental conditions. The observer can then provide audible feedback with the microphone **92** to one or more of the chambers **202-214** using the control **98** to provide one-way communication. The observer in the location **202** can also enable the camera **94** and the speaker **96** so that one or more chambers **202-214** has two-way communication with the observer.

[0062] The control system **200** includes a server **216** that receives signals from one or more of the observer's microphone **92**, camera **94** and speaker **96** then transmits signals from the observer to receiver/transmitters **222, 226, 230, 234, 238** and **242** along hard wired or wireless transmission lines **220, 224, 228, 232, 236** and **240**, respectively. The receiver/transmitters **222, 226, 230, 234, 238** and **242** then transmit signals relating to video and audible stream sent by the observer to the screen and speakers in selected of the chambers **204, 206, 208, 210, 212** and **214** along hard wired or wireless transmission lines **244, 246, 248, 250, 252** and **254**, respectively.

[0063] Similarly, the wearable devices and sensors in the chambers sends signals through the transmission lines **244, 246, 248, 250, 252** and **254**, through the receiver/transmitters **222, 226, 230, 234, 238** and **242**, across transmission lines **220, 224, 228, 232, 236** and **240** to the server **216** and finally to the GUI **100**. Additionally, the observer can manipulate one or more of the modalities **130, 132, 134, 136, 138, 140, 142, 145** and/or **147** based upon the sensed physiological data or the environmental conditions.

[0064] Additionally, the observer can optionally transmit video of the observer that is visible on the screens of the selected chambers **202-214**. The observer can also optionally enable a microphone in a chamber **202-214** such that the user can communicate with the observer.

[0065] As illustrated and described, each of the chambers **204-214** is being utilized. However, if one or more chambers **204-214** is not being utilized, then the system **200** is configured to turn off the communication lines to the unused chamber.

[0066] Referring to FIG. **8**, another control system is illustrated at **300**. The control system **300** includes similar features as that of the control system **200**, besides being in different facilities as illustrated by the line **306**. The control system **300** also includes the concept of multiple observers **302S** and **302B** be able both monitor and control the chambers **302-314** or to divide responsibilities for the chambers.

[0067] Referring to FIG. **9**, another control system is illustrated at **400**. The system **400** allows the observer **402** to interact with a user in a chamber **404** at the user's residence. In some embodiments the observer **402** can be located in a different facility relative to the facility where the chamber **404** is located as illustrated by the line **406**. However, the observer **402** can also be located in the user's residence at the time the user is in the chamber **404**.

[0068] The chamber **404** includes the screen, speakers, microphone and camera as previously discussed. The observer **402** can also be recorded by a camera such that signals relating to the visual and audio recorded by the camera is transmitted along a hard wired or wireless communication line **408** to a receiver/server **410**. The receiver/server **410** sends the signals along a hard wired or wireless communication line **412** to a receiver/router **414** that in turn sends the signal along a hard wired or wireless communication line **416** to the chamber **404**, where the instructor/trainer **402** is streamed to the screen and the audio can be heard through the speakers. The control system **400** also transmits physiological and/or psychological data and environmental conditions that can be monitored by the observer where the observer can optionally manipulate the environmental conditions based upon the physiological and/or psychological data.

[0069] Referring to FIG. **10**, another control system is illustrated at **500**. The control system **500** allows an observer **502** in one facility monitor **503**, communicate with and control environmental conditions in chambers **5** in a second facility **505** and chambers $n \dots n+n$ in a third facility **507**. The number of chambers and number of facilities that can be observed and controlled by the observer **502** can be any selected number, including greater than two.

[0070] The chambers **504, 506, 508, 510, 512, 514** includes the screen, speakers, microphone and camera as previously discussed. The observer **502** is also being recorded by a camera such that signals relating to the visual and audio recorded by the camera is transmitted along a hard wired or wireless communication line **560** to a receiver/server **562**. The receiver/server **562** sends the signals along a hard wired or wireless communication line **564** and **566** to a receiver/router **509** in the facility **505** and the receiver/router **516** in the facility **507** that in turn sends the signal along a hard wired or wireless communication lines **542, 544, 546, 548, 550** and **552** to the chambers **504, 506, 508, 510, 512, 514** through routers/receivers **518, 520, 522, 524, 526** and **528** and lines **530, 532, 534, 536, 538** and **540** all respectively. The control system **50** also transmits physiological and/or psychological data and environmental conditions that can be monitored by the observer where the observer can optionally manipulate the environmental conditions based upon the physiological and/or psychological data.

[0071] Although the present disclosure has been described with reference to preferred embodiments, workers skilled in the art will recognize that changes may be made in form and detail without departing from the spirit and scope of the disclosure.

Claims

1. A method of controlling an environmentally controlled chamber, the method comprising: providing a chamber with at least one infrared heater, a screen configured to stream content, the screen positioned behind an insulating barrier comprising a substantially optically transparent portion, a camera configured to record a user and speakers within the chamber; sending the user's

physiological and/or psychological data and environmental conditions within the chamber to a location remote from the chamber for monitoring; and changing environmental conditions within the chamber based upon the received physiological and/or psychological data or sensed environmental conditions from the location remote from the chamber.

2. The method of claim 1, and further comprising: receiving feedback or instructions over the speakers for the user while within the chamber through the speaker. **3.** The method of claim 1, and further comprising: streaming video content of an observer spaced from the chamber onto the screen for the user to view in real time.

4. The method of claim 1, and further comprising: streaming audio from the microphone to an observer spaced from the chamber such that the observer can receive audio from the user in real time.

5. The method of claim 1, wherein controlling the environmental conditions comprises: adjusting a temperature within the chamber based upon the user's body temperature.

6. The method of claim 5, wherein adjusting the temperature comprises: causing chilled air to be directed into the chamber to lower the temperature within the chamber.

7. The method of claim 5, wherein adjusting the temperature comprises: causing heated air to be directed into the chamber to raise the temperature within the chamber.

8. The method of claim 1, wherein controlling the environmental conditions comprises: adjusting oxygen content within the chamber based upon the user's maximum rate of oxygen consumption.

9. The method of claim 5, wherein controlling the environmental conditions comprises: manipulating a humidity or aroma in the chamber.

10. The method of claim 5, wherein controlling the environmental conditions comprises: adjusting a delivery of electromagnetic radiation or visible light into the chamber.

11. The method of claim 1, wherein the chamber is provided with a recess in a wall with an insulated door, the method further comprising: opening the door to provide access to the recess; positioning a hand held electronic device with the chamber; and closing the door to protect the hand held device from the environmental conditions within the chamber.

12. A method of controlling a plurality of environmentally controlled chambers, the method comprising: providing a plurality of spaced apart chamber with at least one infrared heater, a screen configured to stream content, the screen positioned behind an insulating barrier comprising a substantially optically transparent portion, a camera configured to record a user and speakers within the chamber; sending the user's physiological and/or psychological data and environmental conditions within each of the plurality of chambers to a location remote from the chamber for monitoring; and changing environmental conditions within one of the plurality of chambers based upon the received physiological and/or psychological data or sensed environmental conditions within the one chamber of the plurality of chambers based upon signals sent from the location remote from the one chamber of the plurality of chambers.

13. The method of claim 12, and further comprising: receiving feedback or instructions over the speakers for the user while within one of the plurality of chambers through the speaker.

14. The method of claim 12, and further comprising: streaming video content of an observer spaced from the plurality of chambers onto the screen of at least one of the plurality of chambers for the user to view in real time.

15. The method of claim 12, and further comprising: streaming audio from the microphone to an observer spaced from the chamber such that the observer can receive audio from the user in real time.

16. The method of claim 12, wherein controlling the environmental conditions within one of the plurality of chambers comprises: adjusting a temperature within the one chamber of the plurality of chambers based upon the user's body temperature.

17. The method of claim 16, wherein adjusting the temperature comprises: causing chilled air to be directed into the chamber to lower the temperature within the chamber.

- 18.** The method of claim 16, wherein adjusting the temperature comprises: causing heated air to be directed into the chamber to raise the temperature within the chamber.
- 19.** The method of claim 12, wherein controlling the environmental conditions within one of the plurality of chambers comprises: adjusting oxygen content within the chamber based upon the user's maximum rate of oxygen consumption.
- 20.** The method of claim 12, wherein controlling the environmental conditions within one of the plurality of chambers comprises: adjusting a delivery of electromagnetic radiation or visible light into the chamber.
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